

Digital Forensics Agent (Agent 9)

Deep Technical Specification & Operational Manual

Executive Overview & Purpose

The Forensics Agent conducts deterministic evidence extraction and artifact analysis over the incident data, identifying persistence mechanisms, timelining suspicious activity, and mapping attack artifacts to MITRE ATT&CK; techniques.

1. Architectural Overview & Workflow

- Position in LangGraph Engine: Fourth node in Pipeline B, executing after LogCorrelationAgent.
- Input Consumption: Consumes correlated event clusters, enriched IOCs, and raw payloads.
- State Propagation: Writes state.forensics_report with evidence artifacts, timeline, and persistence findings.

2. Artifact & Persistence Analysis

- Persistence Pattern Detection: Startup keys, cron entries, scheduled tasks, and registry autorun references.
- Process/Service Artifacts: Suspicious binaries, unusual child-process chains, and service registration anomalies.
- File-Hash Attribution: Links discovered hashes to known-bad reputation from the enrichment layer.

3. Incident Timeline Reconstruction

- Chronological Evidence Ledger: Every artifact is stamped with its first-seen event reference.
- Kill-Chain Positioning: Each artifact is mapped to a MITRE ATT&CK; tactic for analyst navigation.
- Chain-of-Custody Fields: Evidence entries include source log reference and detection method for auditability.